

YANG Hao

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EDUCATION

- Northwestern Polytechnical University (NPU) | Xi'an, China** Aug. 2022 - Jul. 2026 (Expected)
- Program: **Carbon Neutrality and Smart Energy Innovation Class (Elite Program, Top 20% high school graduates by province)**, Honors College
 - Major: Computer Science and Technology (Bachelor of Engineering)
 - Minor: Smart Energy
 - GPA: **3.739/4.1**
- University of British Columbia | Kelowna, Canada** Jul. 2023 - Aug. 2023
- Program: Summer Program, Engineering
 - Grade: **Gold Medal**
- University of Oxford | Oxford, United Kingdom** Jul. 2024 - Aug. 2024
- Program: Summer School Program, STEM
 - Grade: **A**

RESEARCH EXPERIENCE

- Research Program on Remote Sensing Super-Resolution** Jul. 2025 - Aug. 2025
Supervised by Prof. Man-On Pun, CUHKSZ
- Outcome: Wrote a paper named **Text-RSSR: Text-Conditioned Remote Sensing Super-Resolution with Cross-Modal Learning**, which is now under revision by IEEE Transactions on Geoscience and Remote Sensing.
 - ✓ Introduced a Text-RSSR model which achieved state-of-the-art performance with PSNR metrics of 16.363 2 dB on Alsat-2B dataset, 27.554 7 dB on UC Merced Land Use Dataset, and 27.400 5 dB on AID.
- Research Program on Remote Sensing Change Detection** Jul. 2024 - Apr. 2025
Supervised by Prof. Zhiyu JIANG, NWPU
- Outcome: Wrote a paper named **Multimodal Difference Augmentation Learning for Remote Sensing Change Detection**, which is now under revision by IEEE Transactions on Geoscience and Remote Sensing.
 - ✓ Introduced an MdaCD model which achieved state-of-the-art performance with IoU scores of 84.88% on LEVIR-CD and 73.13% on SYSU-CD.
 - ✓ Employed a CLIP-Guided Masking process to conceal regions that are highly unlikely to change and subsequently generated text prompts.
 - ✓ Developed a Multimodal Fusion and Difference Augmentation process to enhance the discrepancy between visual and textual information, facilitating differential feature learning and improving the accuracy of change detection.

COMPETITIONS

- The 2021 ICPC China Shaanxi Provincial Programming Contest** Oct. 2022
Competitor, Modeler | Awarded: Bronze Medal
- Background: In a team of three, I was responsible for Problems F and G and collaborated with team members to address Problem B. The Bronze Medal was awarded to the team for completing three Problems (B, G, J).
 - In Problem F, attempted to optimize a brute-force enumeration algorithm but failed to resolve the Problem within the competition time; in Problem G, utilized mathematical modeling to calculate the maximum water storage using the arithmetic series sum formula, successfully solving the Problem; and in Problem B, participated in the process by transforming the complex constraints into a mathematical model and using a Binary Indexed Tree (BIT) for point updates and range queries.
- The 2023 Mathematical Contest in Modeling (MCM)** Feb. 2023
Competitor, Modeler | Awarded: Honorable Mention
- Background: Regularity Behind Random Puzzles, three tasks to be addressed. Completed Tasks 2 and 3, including problem interpretation, algorithm design, code implementation, and contest paper writing.
 - In Task 2, employed a Binary Decision Tree model to analyze the features of the given word and predict the distribution of attempts required to guess the word (1 to 6 attempts or failure); and in Task 3, adopted the K-means++ algorithm to classify words into two categories: "simple" and "difficult," and employed a Binary Logistic Regression model to predict the difficulty of words based on the differences in letter frequency between the two categories.

AWARDS AND TECHNICAL SKILLS

- The 2023 Xiaomi Excellence Award
- The 2023 First Class Scholarship (University-level)
- **Programming: C++, Python, MATLAB, LaTeX, Markdown**
- Qualification: Certified Software Professional (CSP), scored 345/500 (2% among all participants)

LANGUAGES

- Mandarin (Native); English (Fluent, IELTS, GRE); Cantonese (Fluent)